

23. If $f: X \rightarrow Y$ is a map with X and Y homotopy equivalent to CW complexes, show that the pair (M_f, X) is homotopy equivalent to a CW pair, where M_f is the mapping cylinder. Deduce that the mapping cone C_f has the homotopy type of a CW complex.

4.2 Elementary Methods of Calculation

We have not yet computed any nonzero homotopy groups $\pi_n(X)$ with $n \geq 2$. In Chapter 1 the two main tools we used for computing fundamental groups were van Kampen's theorem and covering spaces. In the present section we will study the higher-dimensional analogs of these: the excision theorem for homotopy groups, and fiber bundles. Both of these are quite a bit weaker than their fundamental group analogs, in that they do not directly compute homotopy groups but only give relations between the homotopy groups of different spaces. Their applicability is thus more limited, but suffices for a number of interesting calculations, such as $\pi_n(S^n)$ and more generally the Hurewicz theorem relating the first nonzero homotopy and homology groups of a space. Another noteworthy application is the Freudenthal suspension theorem, which leads to stable homotopy groups and in fact the whole subject of stable homotopy theory.

Excision for Homotopy Groups

What makes homotopy groups so much harder to compute than homology groups is the failure of the excision property. However, there is a certain dimension range, depending on connectivities, in which excision does hold for homotopy groups:

Theorem 4.23. *Let X be a CW complex decomposed as the union of subcomplexes A and B with nonempty connected intersection $C = A \cap B$. If (A, C) is m -connected and (B, C) is n -connected, $m, n \geq 0$, then the map $\pi_i(A, C) \rightarrow \pi_i(X, B)$ induced by inclusion is an isomorphism for $i < m + n$ and a surjection for $i = m + n$.*

This yields the **Freudenthal suspension theorem**:

Corollary 4.24. *The suspension map $\pi_i(S^n) \rightarrow \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n - 1$ and a surjection for $i = 2n - 1$. More generally this holds for the suspension $\pi_i(X) \rightarrow \pi_{i+1}(SX)$ whenever X is an $(n - 1)$ -connected CW complex.*

Proof: Decompose the suspension SX as the union of two cones C_+X and C_-X intersecting in a copy of X . The suspension map is the same as the map

$$\pi_i(X) \approx \pi_{i+1}(C_+X, X) \rightarrow \pi_{i+1}(SX, C_-X) \approx \pi_{i+1}(SX)$$

where the two isomorphisms come from long exact sequences of pairs and the middle map is induced by inclusion. From the long exact sequence of the pair $(C_{\pm}X, X)$ we see that this pair is n -connected if X is $(n-1)$ -connected. The preceding theorem then says that the middle map is an isomorphism for $i+1 < 2n$ and surjective for $i+1 = 2n$. $\square$

Corollary 4.25. $\pi_n(S^n) \approx \mathbb{Z}$, generated by the identity map, for all $n \geq 1$. In particular, the degree map $\pi_n(S^n) \rightarrow \mathbb{Z}$ is an isomorphism.

Proof: From the preceding corollary we know that in the suspension sequence

$$\pi_1(S^1) \rightarrow \pi_2(S^2) \rightarrow \pi_3(S^3) \rightarrow \dots$$

the first map is surjective and all the subsequent maps are isomorphisms. Since $\pi_1(S^1)$ is $\mathbb{Z}$ generated by the identity map, it follows that $\pi_n(S^n)$ for $n \geq 2$ is a finite or infinite cyclic group independent of n , generated by the identity map. The fact that this cyclic group is infinite can be deduced from homology theory since there exist basepoint-preserving maps $S^n \rightarrow S^n$ of arbitrary degree, and degree is a homotopy invariant. Alternatively, if one wants to avoid appealing to homology theory one can use the Hopf bundle $S^1 \rightarrow S^3 \rightarrow S^2$ described in Example 4.45, whose long exact sequence of homotopy groups gives an isomorphism $\pi_1(S^1) \approx \pi_2(S^2)$.

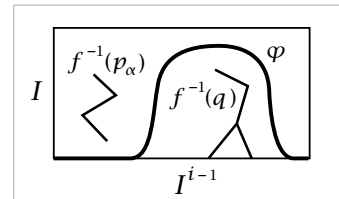
The degree map $\pi_n(S^n) \rightarrow \mathbb{Z}$ is an isomorphism since the map $z \mapsto z^k$ of S^1 has degree k , as do its iterated suspensions by Proposition 2.33. $\square$

Proof of 4.23: We proceed by proving successively more general cases. The first case contains the heart of the argument, and suffices for the calculation of $\pi_n(S^n)$.

Case 1: A is obtained from C by attaching cells e_{α}^{m+1} and B is obtained from C by attaching a cell e^{n+1} . To show surjectivity of $\pi_i(A, C) \rightarrow \pi_i(X, B)$ we start with a map $f: (I^i, \partial I^i, J^{i-1}) \rightarrow (X, B, x_0)$. The image of f is compact and therefore meets only finitely many of these cells e_{α}^{m+1} and e^{n+1} . By repeated applications of Lemma 4.10 we may homotope f , through maps $(I^i, \partial I^i, J^{i-1}) \rightarrow (X, B, x_0)$, so that the preimages $f^{-1}(\Delta_{\alpha}^{m+1})$ and $f^{-1}(\Delta^{n+1})$ of simplices in e_{α}^{m+1} and e^{n+1} are finite unions of convex polyhedra, on each of which f is the restriction of a linear surjection from $\mathbb{R}^i$ onto $\mathbb{R}^{m+1}$ or $\mathbb{R}^{n+1}$.

Claim: If $i \leq m+n$, then there exist points $p_{\alpha} \in \Delta_{\alpha}^{m+1}$, $q \in \Delta^{n+1}$, and a map $\varphi: I^{i-1} \rightarrow [0, 1)$ such that:

- (a) $f^{-1}(q)$ lies below the graph of φ in $I^{i-1} \times I = I^i$.
- (b) $f^{-1}(p_{\alpha})$ lies above the graph of φ for each α .
- (c) $\varphi = 0$ on ∂I^{i-1} .



Granting this, let f_t be a homotopy of f excising the region under the graph of φ by restricting f to the region above the graph of $t\varphi$ for $0 \leq t \leq 1$. By (b), $f_t(I^{i-1})$ is disjoint from $P = \bigcup_{\alpha} \{p_{\alpha}\}$ for all t , and by (a), $f_1(I^i)$ is disjoint from $Q = \{q\}$. This

means that in the commutative diagram at the right the given element $[f]$ in the upper-right group, when regarded as an element of the lower-right group,

$$\begin{array}{ccc} \pi_i(A, C) & \longrightarrow & \pi_i(X, B) \\ \downarrow \approx & & \downarrow \approx \\ \pi_i(X - Q, X - Q - P) & \longrightarrow & \pi_i(X, X - P) \end{array}$$

is equal to the element $[f_1]$ in the image of the lower horizontal map. Since the vertical maps are isomorphisms, this proves the surjectivity statement.

Now we prove the Claim. For any $q \in \Delta^{n+1}$, $f^{-1}(q)$ is a finite union of convex polyhedra of dimension $\leq i - n - 1$ since $f^{-1}(\Delta^{n+1})$ is a finite union of convex polyhedra on each of which f is the restriction of a linear surjection $\mathbb{R}^i \rightarrow \mathbb{R}^{n+1}$. We wish to choose the points $p_\alpha \in \Delta_\alpha^{m+1}$ so that not only is $f^{-1}(q)$ disjoint from $f^{-1}(p_\alpha)$ for each α , but also so that $f^{-1}(q)$ and $f^{-1}(p_\alpha)$ have disjoint images under the projection $\pi: I^i \rightarrow I^{i-1}$. This is equivalent to saying that $f^{-1}(p_\alpha)$ is disjoint from $T = \pi^{-1}(\pi(f^{-1}(q)))$, the union of all segments $\{x\} \times I$ meeting $f^{-1}(q)$. This set T is a finite union of convex polyhedra of dimension $\leq i - n$ since $f^{-1}(q)$ is a finite union of convex polyhedra of dimension $\leq i - n - 1$. Since linear maps cannot increase dimension, $f(T) \cap \Delta_\alpha^{m+1}$ is also a finite union of convex polyhedra of dimension $\leq i - n$. Thus if $m + 1 > i - n$, there is a point $p_\alpha \in \Delta_\alpha^{m+1}$ not in $f(T)$. This gives $f^{-1}(p_\alpha) \cap T = \emptyset$ if $i \leq m + n$. Hence we can choose a neighborhood U of $\pi(f^{-1}(q))$ in I^{n-1} disjoint from $\pi(f^{-1}(p_\alpha))$ for all α . Then there exists $\varphi: I^{i-1} \rightarrow [0, 1]$ having support in U , with $f^{-1}(q)$ lying under the graph of φ . This verifies the Claim, and so finishes the proof of surjectivity in Case 1.

For injectivity in Case 1 the argument is very similar. Suppose we have two maps $f_0, f_1: (I^i, \partial I^i, J^{i-1}) \rightarrow (A, C, x_0)$ representing elements of $\pi_i(A, C, x_0)$ having the same image in $\pi_i(X, B, x_0)$. Thus there is a homotopy from f_0 to f_1 in the form of a map $F: (I^i, \partial I^i, J^{i-1}) \times [0, 1] \rightarrow (X, B, x_0)$. After a preliminary deformation of F via Lemma 4.10, we construct a function $\varphi: I^{i-1} \times I \rightarrow [0, 1]$ separating $F^{-1}(q)$ from the sets $F^{-1}(p_\alpha)$ as before. This allows us to excise $F^{-1}(q)$ from the domain of F , from which it follows that f_0 and f_1 represent the same element of $\pi_i(A, C, x_0)$. Since $I^i \times I$ now plays the role of I^i , the dimension i is replaced by $i + 1$ and the dimension restriction $i \leq m + n$ becomes $i + 1 \leq m + n$, or $i < m + n$.

Case 2: A is obtained from C by attaching $(m + 1)$ -cells as in Case 1 and B is obtained from C by attaching cells of dimension $\geq n + 1$. To show surjectivity of $\pi_i(A, C) \rightarrow \pi_i(X, B)$, consider a map $f: (I^i, \partial I^i, J^{i-1}) \rightarrow (X, B, x_0)$ representing an element of $\pi_i(X, B)$. The image of f is compact, meeting only finitely many cells, and by repeated applications of Case 1 we can push f off the cells of $B - C$ one at a time, in order of decreasing dimension. Injectivity is quite similar, starting with a homotopy $F: (I^i, \partial I^i, J^{i-1}) \times [0, 1] \rightarrow (X, B, x_0)$ and pushing this off cells of $B - C$.

Case 3: A is obtained from C by attaching cells of dimension $\geq m + 1$ and B is as in Case 2. We may assume all cells of $A - C$ have dimension $\leq m + n + 1$ since higher-dimensional cells have no effect on π_i for $i \leq m + n$, by cellular approximation. Let

$A_k \subset A$ be the union of C with the cells of A of dimension $\leq k$ and let $X_k = A_k \cup B$. We prove the result for $\pi_i(A_k, C) \rightarrow \pi_i(X_k, B)$ by induction on k . The induction starts with $k = m + 1$, which is Case 2. For the induction step consider the following commutative diagram formed by the exact sequences of the triples (A_k, A_{k-1}, C) and (X_k, X_{k-1}, B) :

$$\begin{array}{ccccccccc} \pi_{i+1}(A_k, A_{k-1}) & \longrightarrow & \pi_i(A_{k-1}, C) & \longrightarrow & \pi_i(A_k, C) & \longrightarrow & \pi_i(A_k, A_{k-1}) & \longrightarrow & \pi_{i-1}(A_{k-1}, C) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \pi_{i+1}(X_k, X_{k-1}) & \longrightarrow & \pi_i(X_{k-1}, B) & \longrightarrow & \pi_i(X_k, B) & \longrightarrow & \pi_i(X_k, X_{k-1}) & \longrightarrow & \pi_{i-1}(X_{k-1}, B) \end{array}$$

When $i < m + n$ the first and fourth vertical maps are isomorphisms by Case 2, while by induction the second and fifth maps are isomorphisms, so the middle map is an isomorphism by the five-lemma. Similarly, when $i = m + n$ the second and fourth maps are surjective and the fifth map is injective, which is enough to imply the middle map is surjective by one half of the five-lemma. When $i = 2$ the diagram may contain nonabelian groups and the two terms on the right may not be groups, but the five-lemma remains valid in this generality, with trivial modifications to the proof in §2.1. When $i = 1$ the assertion about $\pi_1(A, C) \rightarrow \pi_1(X, B)$ follows by a direct argument: If $m \geq 1$ then both terms are trivial, while if $m = 0$ then $n \geq 1$ and the result follows by cellular approximation.

After these special cases we can now easily deal with the general case. The connectivity assumptions on the pairs (A, C) and (B, C) imply by Corollary 4.16 that they are homotopy equivalent to pairs (A', C) and (B', C) as in Case 3, via homotopy equivalences fixed on C , so these homotopy equivalences fit together to give a homotopy equivalence $A \cup B \simeq A' \cup B'$. Thus the general case reduces to Case 3. $\square$

Example 4.26. The calculation of $\pi_n(S^n)$ can be extended to show that $\pi_n(\bigvee_{\alpha} S_{\alpha}^n)$ for $n \geq 2$ is free abelian with basis the homotopy classes of the inclusions $S_{\alpha}^n \hookrightarrow \bigvee_{\alpha} S_{\alpha}^n$. Suppose first that there are only finitely many summands S_{α}^n . We can regard $\bigvee_{\alpha} S_{\alpha}^n$ as the n -skeleton of the product $\prod_{\alpha} S_{\alpha}^n$, where S_{α}^n is given its usual CW structure and $\prod_{\alpha} S_{\alpha}^n$ has the product CW structure. Since $\prod_{\alpha} S_{\alpha}^n$ has cells only in dimensions a multiple of n , the pair $(\prod_{\alpha} S_{\alpha}^n, \bigvee_{\alpha} S_{\alpha}^n)$ is $(2n - 1)$ -connected. Hence from the long exact sequence of homotopy groups for this pair we see that the inclusion $\bigvee_{\alpha} S_{\alpha}^n \hookrightarrow \prod_{\alpha} S_{\alpha}^n$ induces an isomorphism on π_n if $n \geq 2$. By Proposition 4.2 we have $\pi_n(\prod_{\alpha} S_{\alpha}^n) \approx \bigoplus_{\alpha} \pi_n(S_{\alpha}^n)$, a free abelian group with basis the inclusions $S_{\alpha}^n \hookrightarrow \prod_{\alpha} S_{\alpha}^n$, so the same is true for $\bigvee_{\alpha} S_{\alpha}^n$. This takes care of the case of finitely many S_{α}^n 's.

To reduce the case of infinitely many summands S_{α}^n to the finite case, consider the homomorphism $\Phi: \bigoplus_{\alpha} \pi_n(S_{\alpha}^n) \rightarrow \pi_n(\bigvee_{\alpha} S_{\alpha}^n)$ induced by the inclusions $S_{\alpha}^n \hookrightarrow \bigvee_{\alpha} S_{\alpha}^n$. Then Φ is surjective since any map $f: S^n \rightarrow \bigvee_{\alpha} S_{\alpha}^n$ has compact image contained in the wedge sum of finitely many S_{α}^n 's, so by the finite case already proved, $[f]$ is in the image of Φ . Similarly, a nullhomotopy of f has compact image contained in a finite wedge sum of S_{α}^n 's, so the finite case also implies that Φ is injective.

Example 4.27. Let us show that $\pi_n(S^1 \vee S^n)$ for $n \geq 2$ is free abelian on a countably infinite number of generators. By Proposition 4.1 we may compute $\pi_i(S^1 \vee S^n)$ for $i \geq 2$ by passing to the universal cover. This consists of a copy of $\mathbb{R}$ with a sphere S_k^n attached at each integer point $k \in \mathbb{R}$, so it is homotopy equivalent to $\bigvee_k S_k^n$. The preceding Example 4.26 says that $\pi_n(\bigvee_k S_k^n)$ is free abelian with basis represented by the inclusions of the wedge summands. So a basis for π_n of the universal cover of $S^1 \vee S^n$ is represented by maps that lift the maps obtained from the inclusion $S^n \hookrightarrow S^1 \vee S^n$ by the action of the various elements of $\pi_1(S^1 \vee S^n) \approx \mathbb{Z}$. This means that $\pi_n(S^1 \vee S^n)$ is a free $\mathbb{Z}[\pi_1(S^1 \vee S^n)]$ -module on a single basis element, the homotopy class of the inclusion $S^n \hookrightarrow S^1 \vee S^n$. Writing a generator of $\pi_1(S^1 \vee S^n)$ as t , the group ring $\mathbb{Z}[\pi_1(S^1 \vee S^n)]$ becomes $\mathbb{Z}[t, t^{-1}]$, the Laurent polynomials in t and t^{-1} with $\mathbb{Z}$ coefficients, and we have $\pi_n(S^1 \vee S^n) \approx \mathbb{Z}[t, t^{-1}]$.

This example shows that the homotopy groups of a finite CW complex need not be finitely generated, in contrast to the homology groups. However, if we restrict attention to spaces with trivial action of π_1 on all π_n 's, then a theorem of Serre, proved in [SSAT], says that the homotopy groups of such a space are finitely generated iff the homology groups are finitely generated.

In this example, $\pi_n(S^1 \vee S^n)$ is finitely generated as a $\mathbb{Z}[\pi_1]$ -module, but there are finite CW complexes where even this fails. This happens in fact for $\pi_3(S^1 \vee S^2)$, according to Exercise 38 at the end of this section. In §4.A we construct more complicated examples for each π_n with $n > 1$, in particular for π_2 .

A useful tool for more complicated calculations is the following general result:

Proposition 4.28. *If a CW pair (X, A) is r -connected and A is s -connected, with $r, s \geq 0$, then the map $\pi_i(X, A) \rightarrow \pi_i(X/A)$ induced by the quotient map $X \rightarrow X/A$ is an isomorphism for $i \leq r + s$ and a surjection for $i = r + s + 1$.*

Proof: Consider $X \cup CA$, the complex obtained from X by attaching a cone CA along $A \subset X$. Since CA is a contractible subcomplex of $X \cup CA$, the quotient map $X \cup CA \rightarrow (X \cup CA)/CA = X/A$ is a homotopy equivalence by Proposition 0.17. So we have a commutative diagram

$$\begin{array}{ccccc} \pi_i(X, A) & \longrightarrow & \pi_i(X \cup CA, CA) & \longrightarrow & \pi_i(X \cup CA/CA) = \pi_i(X/A) \\ & & \uparrow \approx & \nearrow \approx & \\ & & \pi_i(X \cup CA) & & \end{array}$$

where the vertical isomorphism comes from a long exact sequence. Now apply the excision theorem to the first map in the diagram, using the fact that (CA, A) is $(s + 1)$ -connected if A is s -connected, which comes from the exact sequence for the pair (CA, A) . $\square$

Example 4.29. Suppose X is obtained from a wedge of spheres $\bigvee_\alpha S_\alpha^n$ by attaching cells e_β^{n+1} via basepoint-preserving maps $\varphi_\beta: S^n \rightarrow \bigvee_\alpha S_\alpha^n$, with $n \geq 2$. By cellular

approximation we know that $\pi_i(X) = 0$ for $i < n$, and we shall show that $\pi_n(X)$ is the quotient of the free abelian group $\pi_n(\bigvee_\alpha S_\alpha^n) \approx \bigoplus_\alpha \mathbb{Z}$ by the subgroup generated by the classes $[\varphi_\beta]$. Any subgroup can be realized in this way, by choosing maps φ_β to represent a set of generators for the subgroup, so it follows that every abelian group can be realized as $\pi_n(X)$ for such a space $X = (\bigvee_\alpha S_\alpha^n) \cup_\beta e_\beta^{n+1}$. This is the higher-dimensional analog of the construction in Corollary 1.28 of a 2-dimensional CW complex with prescribed fundamental group.

To see that $\pi_n(X)$ is as claimed, consider the following portion of the long exact sequence of the pair $(X, \bigvee_\alpha S_\alpha^n)$:

$$\pi_{n+1}(X, \bigvee_\alpha S_\alpha^n) \xrightarrow{\partial} \pi_n(\bigvee_\alpha S_\alpha^n) \longrightarrow \pi_n(X) \longrightarrow 0$$

The quotient $X/\bigvee_\alpha S_\alpha^n$ is a wedge of spheres S_β^{n+1} , so the preceding proposition and Example 4.26 imply that $\pi_{n+1}(X, \bigvee_\alpha S_\alpha^n)$ is free with basis the characteristic maps of the cells e_β^{n+1} . The boundary map ∂ takes these to the classes $[\varphi_\beta]$, and the result follows.

Eilenberg-MacLane Spaces

A space X having just one nontrivial homotopy group $\pi_n(X) \approx G$ is called an **Eilenberg-MacLane space** $K(G, n)$. The case $n = 1$ was considered in §1.B, where the condition that $\pi_i(X) = 0$ for $i > 1$ was replaced by the condition that X have a contractible universal cover, which is equivalent for spaces that have a universal cover of the homotopy type of a CW complex.

We can build a CW complex $K(G, n)$ for arbitrary G and n , assuming G is abelian if $n > 1$, in the following way. To begin, let X be an $(n-1)$ -connected CW complex of dimension $n+1$ such that $\pi_n(X) \approx G$, as was constructed in Example 4.29 above when $n > 1$ and in Corollary 1.28 when $n = 1$. Then we showed in Example 4.17 how to attach higher-dimensional cells to X to make π_i trivial for $i > n$ without affecting π_n or the lower homotopy groups.

By taking products of $K(G, n)$'s for varying n we can then realize any sequence of groups G_n , abelian for $n > 1$, as the homotopy groups π_n of a space.

A fair number of $K(G, 1)$'s arise naturally in a variety of contexts, and a few of these are mentioned in §1.B. By contrast, naturally occurring $K(G, n)$'s for $n \geq 2$ are rare. It seems the only real example is $\mathbb{CP}^\infty$, which is a $K(\mathbb{Z}, 2)$ as we shall see in Example 4.50. One could of course trivially generalize this example by taking a product of $\mathbb{CP}^\infty$'s to get a $K(G, 2)$ with G a product of $\mathbb{Z}$'s.

Actually there is a fairly natural construction of a $K(\mathbb{Z}, n)$ for arbitrary n , the infinite symmetric product $SP(S^n)$ defined in §3.C. In §4.K we prove that the functor SP has the surprising property of converting homology groups into homotopy groups, namely $\pi_i(SP(X)) \approx H_i(X; \mathbb{Z})$ for all $i > 0$ and all connected CW complexes X . Taking X to be a sphere, we deduce that $SP(S^n)$ is a $K(\mathbb{Z}, n)$. More generally, $SP(M(G, n))$ is a $K(G, n)$ for each Moore space $M(G, n)$.

Having shown the existence of $K(G, n)$'s, we now consider the uniqueness question, which has the nicest possible answer:

Proposition 4.30. *The homotopy type of a CW complex $K(G, n)$ is uniquely determined by G and n .*

The proof will be based on a more technical statement:

Lemma 4.31. *Let X be a CW complex of the form $(\bigvee_{\alpha} S_{\alpha}^n) \cup_{\beta} e_{\beta}^{n+1}$ for some $n \geq 1$. Then for every homomorphism $\psi: \pi_n(X) \rightarrow \pi_n(Y)$ with Y path-connected there exists a map $f: X \rightarrow Y$ with $f_* = \psi$.*

Proof: To begin, let f send the natural basepoint of $\bigvee_{\alpha} S_{\alpha}^n$ to a chosen basepoint $y_0 \in Y$. Extend f over each sphere S_{α}^n via a map representing $\psi([i_{\alpha}])$ where i_{α} is the inclusion $S_{\alpha}^n \hookrightarrow X$. Thus for the map $f: X^n \rightarrow Y$ constructed so far we have $f_*([i_{\alpha}]) = \psi([i_{\alpha}])$ for all α , hence $f_*([\varphi]) = \psi([\varphi])$ for all basepoint-preserving maps $\varphi: S^n \rightarrow X^n$ since the i_{α} 's generate $\pi_n(X^n)$. To extend f over a cell e_{β}^{n+1} all we need is that the composition of the attaching map $\varphi_{\beta}: S^n \rightarrow X^n$ for this cell with f be nullhomotopic in Y . But this composition $f\varphi_{\beta}$ represents $f_*([\varphi_{\beta}]) = \psi([\varphi_{\beta}])$, and $\psi([\varphi_{\beta}]) = 0$ because $[\varphi_{\beta}]$ is zero in $\pi_n(X)$ since φ_{β} is nullhomotopic in X via the characteristic map of e_{β}^{n+1} . Thus we obtain an extension $f: X \rightarrow Y$. This has $f_* = \psi$ since the elements $[i_{\alpha}]$ generate $\pi_n(X^n)$ and hence also $\pi_n(X)$ by cellular approximation. $\square$

Proof of 4.30: Suppose K and K' are $K(G, n)$ CW complexes. Since homotopy equivalence is an equivalence relation, there is no loss of generality if we assume K is a particular $K(G, n)$, namely one constructed from a space X as in the lemma by attaching cells of dimension $n+2$ and greater. By the lemma there is a map $f: X \rightarrow K'$ inducing an isomorphism on π_n . To extend this f over K we proceed inductively. For each cell e^{n+2} , the composition of its attaching map with f is nullhomotopic in K' since $\pi_{n+1}(K') = 0$, so f extends over this cell. The same argument applies for all the higher-dimensional cells in turn. The resulting $f: K \rightarrow K'$ is a homotopy equivalence since it induces isomorphisms on all homotopy groups. $\square$

The Hurewicz Theorem

Using the calculations of homotopy groups done above we can easily prove the simplest and most often used cases of the Hurewicz theorem:

Theorem 4.32. *If a space X is $(n-1)$ -connected, $n \geq 2$, then $\tilde{H}_i(X) = 0$ for $i < n$ and $\pi_n(X) \approx H_n(X)$. If a pair (X, A) is $(n-1)$ -connected, $n \geq 2$, with A simply-connected and nonempty, then $H_i(X, A) = 0$ for $i < n$ and $\pi_n(X, A) \approx H_n(X, A)$.*

Thus the first nonzero homotopy and homology groups of a simply-connected space occur in the same dimension and are isomorphic. One cannot expect any nice